GREEN'S vs STOKES' THEOREM

Green's circulation theorem is Stokes on a plane.

GREEN'S THEOREM

Planar region D in the xy-plane [R Green Plot Here]

$$\oint_C (P\,\mathrm{d}x + Q\,\mathrm{d}y) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}A$$

Scalar curl integrated over a planar region.

STOKES' THEOREM

Surface S bounded by closed curve C [R Stokes Plot Here]

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \iint_S (\nabla \times \boldsymbol{F}) \cdot \mathrm{d}\boldsymbol{S}$$

Curl flux over a surface.

Recover Green from Stokes

$\boldsymbol{F} = (P, Q, 0)$	$\widehat{\boldsymbol{n}} = (0, 0, 1)$	$\boldsymbol{F} \cdot \mathrm{d}r = P\mathrm{d}x + Q\mathrm{d}y$	$(\nabla \times \boldsymbol{F}) \cdot \widehat{\boldsymbol{n}}$	$\mathrm{d}S = \mathrm{d}A$

On a flat xy-surface, Stokes reduces exactly to Green's circulation form.